Click Secure : Malicious URL & message detection

Group Id: PR001430

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Abstract

With the rise of digital communication, cyber threats like spam messages and malicious URLs have become a major concern, leading to phishing attacks, data theft, and financial fraud. Traditional detection methods often struggle to keep up with evolving threats, making machine learning-based solutions essential. **Click Secure** is an AI-powered platform designed to detect spam messages and malicious URLs in real-time. Using SVM and Random Forest algorithms, it accurately classifies messages and links as safe or harmful, achieving 96.43% accuracy. Built with Python, Django, and REST APIs, the system provides a user-friendly interface for quick and reliable threat detection. By leveraging advanced classification techniques, Click Secure enhances online security, protects against phishing attacks, and ensures a safer browsing experience for users. Whether for personal or professional use, it serves as an efficient and dependable cybersecurity tool.

93.41%

Introduction

With the rapid growth of digital communication, cyber threats such as spam messages and malicious URLs have become a significant concern. Hackers and scammers use deceptive tactics to trick users into clicking harmful links, leading to phishing attacks, financial fraud, and data breaches. Traditional spam detection methods often fail to keep up with evolving threats, making advanced solutions essential . Click Secure is an innovative online spam detection platform designed to identify and block harmful messages and URLs using machine learning algorithms. The system offers a user-friendly interface where users can easily input a message or link and receive instant results on its safety. By leveraging advanced detection techniques, Click Secure ensures high accuracy while minimizing false positives .This project aims to provide a reliable, real-time, and efficient solution to protect individuals and businesses from cyber threats. It not only enhances online safety but also contributes to a secure digital environment. With the rise in phishing scams and malware attacks, Click Secure serves as a powerful tool to safeguard users from potential risks. By integrating cutting-edge AI techniques, this system empowers users to navigate the internet with confidence, ensuring a safer browsing experience.

Table 1. Model Performance Overview.

Model Name	Description	Accuracy (%)
Random Forest	Decision tree-based model that classifies based on multiple trees.	94.31%
SVM Model	Support Vector Machine that classifies using hyperplanes.	96.43%

Methods and Materials

For this project, we used the Spam.tsv dataset, which contains both spam (unwanted messages) and ham (regular messages). This dataset helped train the model to classify messages accurately. We developed the system using Python (Anaconda) and Django, with the Django REST Framework for API integration. Machine learning libraries such as Pandas, NumPy, Scikit-learn, and Matplotlib were used for data processing, model training, and visualization. Development was done in VS Code. The process began with data preprocessing, where we cleaned and balanced the dataset to improve accuracy. TF-IDF Vectorization was applied to convert text into numerical format for better analysis. We trained two machine learning models: Support Vector Machine (SVM) and Random Forest, comparing their performance. After testing, the SVM model achieved 96.43% accuracy, outperforming Random Forest. The final model was deployed using Django, making it accessible through a web-based API that allows users to input messages and URLs for real-time spam and malicious link detection. This system ensures a high level of security by identifying harmful content and preventing phishing attacks effectively.

Results

The Click Secure system has demonstrated exceptional performance in detecting spam messages and malicious URLs. In terms of model accuracy, the SVM model outperformed the Random Forest model, achieving an impressive **96.43%** accuracy. The Random Forest model, while effective, achieved a slightly lower accuracy of **94.31%**. This difference in performance highlights the strength of the SVM model in detecting malicious content. The system's ability to minimize false positives and offer quick, real-time classification makes it an effective tool for ensuring secure online communication. These results confirm that Click Secure is a highly accurate solution for spam and malicious URL detection, capable of protecting users against evolving cyber threats.

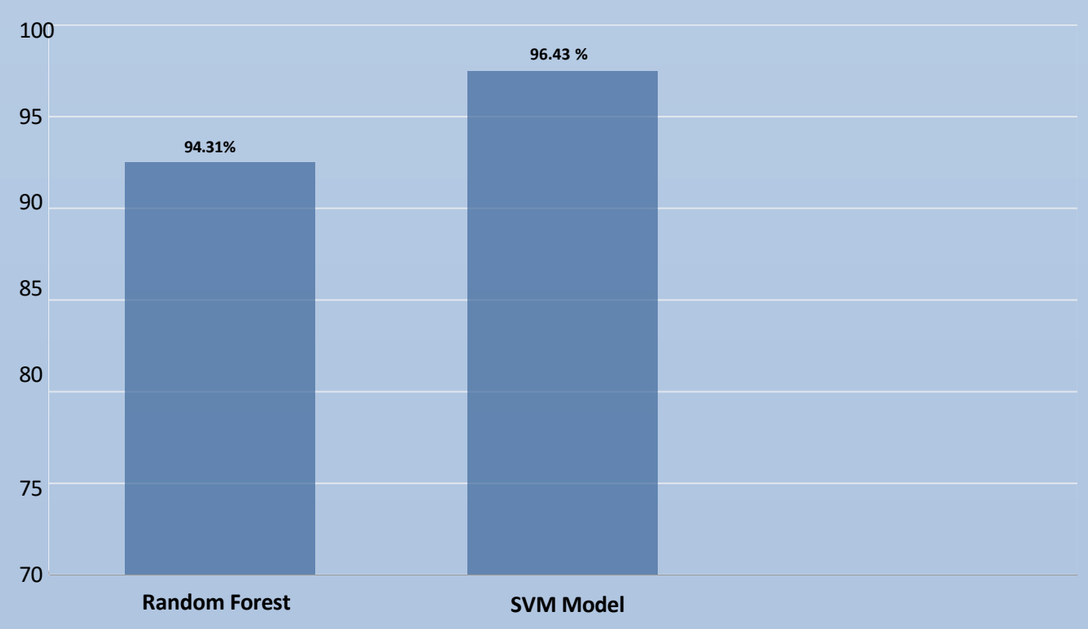

Chart 1 . Model Accuracy Comparison.

Discussion

Click Secure efficiently identifies spam messages and malicious URLs through the content and pattern analysis of messages and links. SVM provided greater accuracy over Random Forest, demonstrating its superiority in handling spam detection. Despite overall efficiency, it continues to be a challenge in the detection of new and advanced spam and phishing attacks. Future enhancements can involve incorporating deep learning models for precise predictions and sophisticated URL scanning methods to address emerging threats. In the face of these challenges, Click Secure is a crucial step forward toward enhanced cybersecurity and blocking online fraud.

Fig 1 : Stay Safe, Click Secure.

Fig 2 : Shielding You from Malicious Threats.

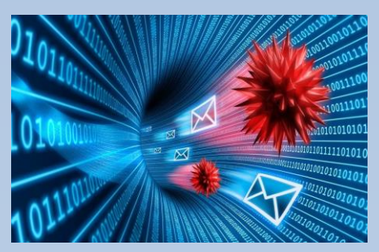

Fig 3 : Stop Spam Before It Stops You.

Conclusions

Click Secure is an AI-driven security tool designed to combat cyber threats, offering a highly accurate and real-time spam and malicious URL detection system. With an accuracy of 96.43%, the system enhances digital security and helps prevent financial fraud and data breaches. Moving forward, the project will focus on further refining the system by incorporating deep learning techniques and expanding the URL threat databases to improve detection accuracy and keep pace with evolving cyber threats.

Acknowledgment

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